
An LLM Judge Loses Reliability on Problems in Its Training Corpus, but Only When the Reference Answer Is Withheld

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Abstract

Automated evaluators are trained artifacts, so a verdict on a benchmark item may depend on whether the evaluator itself read that item during training, yet the corpora of models normally deployed as judges are closed and the question is usually left open. Instella publishes its complete pretraining mixture, and its stage two component derives from the GSM8K training split and not the test split, so membership is fixed by exact thirteen gram containment instead of estimated: the adopted threshold flags none of 1000 known negative test items, and the looser any match rule of Brown et al. flags 3, a false positive rate of 0.3 percent. Difficulty matched arms of 250 seen and 250 unseen problems expand into 2017 candidate solutions produced by four Instella checkpoints, and three judges spanning a capability range, Instella 3B Instruct, Qwen2.5 7B Instruct and Qwen2.5 14B Instruct, return 48,408 verdicts at temperature zero under two prompts that differ only in whether the gold answer is supplied. Chance corrected agreement is shown to be the wrong statistic here and is reported as a cautionary result: Cohen’s kappa manufactures a seen minus unseen gap of -0.292 $[-0.356, -0.228]$ purely from a base rate difference of 77 against 53 percent, a textbook instance of the Feinstein and Cicchetti paradox, while balanced accuracy on the identical verdicts puts the same cell at $+0.054$. Under balanced accuracy a dissociation appears. Supplying the reference answer drives the membership gap toward zero as judges improve, running -0.054 , -0.036 and -0.023 , the last spanning zero, and toward zero more sharply still on a second generator at -0.122 , -0.002 and $+0.008$. Withholding it removes the convergence entirely, at -0.081 , -0.200 and -0.184 , with the effect emerging exactly where a judge first becomes competent, since control condition balanced accuracy climbs from 0.514 to 0.937 to 0.946 across the ladder. Scale does not repair the reference free gap, which rules out a weak judge artifact. Abstention stays between 0.0 and 2.6 percent throughout, so selective non response does not generate the pattern. All intervals are cluster bootstraps over parent problems against a measured intraclass correlation of 0.48, since resampling rows would contract them by a design effect near 2.4. The practical consequence is narrow and testable: reference free LLM grading, the configuration used whenever no gold answer exists, degrades on exactly the items most likely to sit in a judge’s training data, and supplying a reference restores it.

1 Introduction

Using a language model to grade another model’s output rests on an assumption that is seldom examined, that the evaluator’s reliability does not depend on the material being evaluated. If a judge has memorized a benchmark problem, its verdict on a candidate solution to that problem may reflect what it remembers instead of what it was shown. The verdict is then a measurement whose

38 calibration varies with the thing being measured, which is the property an instrument is supposed
39 not to have.

40 Testing the assumption needs the judge’s training corpus. For the closed weight models usually de-
41 ployed as judges that is unavailable, so existing audits study properties observable from the outside.
42 Zheng et al. [Zheng et al., 2023] establish the paradigm and document position bias, verbosity bias
43 and self preference, and report that strong judges agree with human preference above 80 percent of
44 the time. None of that work can condition on membership, because membership is not knowable for
45 the judges in question.

46 Instella [Liu et al., 2025] makes it knowable for one judge and one corpus. AMD publishes the
47 full pretraining mixture, and the stage two synthetic mathematics corpus derives from the GSM8K
48 training split [Cobbe et al., 2021] and not from the test split. Membership of a problem in that
49 corpus follows from exact thirteen gram containment [Brown et al., 2020]. Since the corpus is
50 generated only from train, every test item is a known negative, which converts detector error from
51 an assumption into a measured quantity.

52 That gives an item partition with verified provenance, and the study puts the same items to three
53 judges under two prompts differing in one respect, whether the gold answer for the graded variant
54 appears. The reference based condition reduces grading to comparison, so a remembered answer
55 buys the judge nothing. The reference free condition requires assessing the presented reasoning,
56 which is where recall could substitute for verification. Because both conditions run over identical
57 items and identical candidate solutions, the contrast between them is internal to each judge and does
58 not depend on what the arm label means for that judge’s own corpus, a point Section 6 returns to.

59 Three contributions follow. First, judge reliability measured against membership fixed by contain-
60 ment rather than inferred, over 48,408 verdicts. Second, a dissociation in which supplying a refer-
61 ence answer removes the membership gap while withholding it does not, with the gap appearing
62 precisely where judges become competent. Third, a documented case of the kappa paradox produc-
63 ing a large and interval excluding effect on data where the effect is absent, which is worth reporting
64 because chance corrected agreement remains a common choice for exactly this measurement.

65 2 Design

66 **Verified arms.** Item text is lowercased, reduced to alphanumeric tokens, and cut into every win-
67 dows of 13 consecutive tokens. Containment is the fraction of an item’s windows appearing in the
68 corpus. Because the corpus derives only from GSM8K train, the test split calibrates the rule: the
69 any match convention of Brown et al. [Brown et al., 2020] flags 3 of 1000 known negatives, and
70 the threshold adopted here flags none of them. Identifiers are namespaced by split, without which
71 train and test items collide and containment exceeds unity. From the eligible pools, 250 seen and
72 250 unseen parent problems are drawn matched on reasoning step count, so arm composition cannot
73 stand in for a membership effect.

74 **Candidate solutions.** Each parent expands into one original and four variants, two answer pre-
75 serving and two answer changing, giving 2017 rows. Four Instella checkpoints generate candidate
76 solutions under pinned decoding, and exact match against ground truth supplies the label each judge
77 is scored against. Two thirds of graded rows are variants, and the answer changing variants carry
78 different correct answers from their originals, which matters for the mechanism in Section 5.

79 **Judges and conditions.** Three judges span a deliberate capability range: Instella 3B Instruct,
80 Qwen2.5 7B Instruct and Qwen2.5 14B Instruct. Each emits one word, correct or incorrect, at
81 temperature zero under a 24 token budget. The two conditions differ in one element of the prompt.
82 Reference based grading supplies the gold answer for the variant at hand and is the control, since
83 recall of a remembered solution cannot help once the solution is given. Reference free grading
84 withholds it.

85 **Metric and inference.** Balanced accuracy, the unweighted mean of sensitivity and specificity, is
86 used throughout. Both components condition on the true class, so neither moves with the base rate,
87 and their mean stays comparable across arms whose ground truth composition differs. Chance is
88 0.5. Section 3.1 shows why the obvious alternative fails here. Intervals are bootstraps over parent

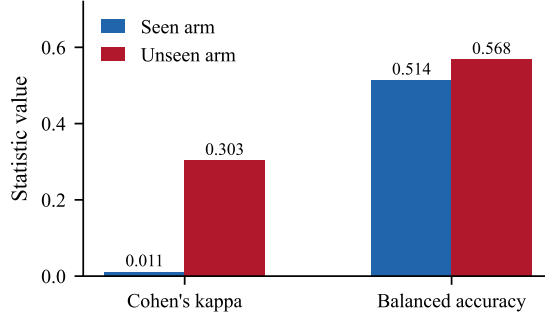


Figure 1: Two statistics computed on one set of verdicts, Instella 3B Instruct grading stage two outputs in the reference free condition. Cohen’s kappa reports a large seen minus unseen gap that reflects a ground truth base rate difference of 77 against 53 percent between the arms. Balanced accuracy, which conditions on the true class, does not.

Table 1: Balanced accuracy, seen minus unseen, grading instruct outputs. Intervals are 95 percent bootstraps over parent problems, 4000 draws. An asterisk marks an interval excluding zero.

Judge	Reference given (control)	Reference withheld
Instella 3B Instruct	−0.054 [−0.083, −0.026]*	−0.081 [−0.109, −0.054]*
Qwen2.5 7B Instruct	−0.036 [−0.068, −0.008]*	−0.200 [−0.248, −0.149]*
Qwen2.5 14B Instruct	−0.023 [−0.057, +0.006]	−0.184 [−0.231, −0.133]*

89 problems rather than rows, because the five rows of one problem are not five independent observa-
90 tions; the measured intraclass correlation is approximately 0.48, and resampling rows would contract
91 intervals by a design effect near 2.4.

92 3 Results

93 3.1 Chance corrected agreement is the wrong statistic here

94 Judge accuracy was first measured as Cohen’s kappa against exact match ground truth, and the result
95 looked conclusive. Grading stage two outputs, the seen minus unseen difference came to −0.292 on
96 an interval of [−0.356, −0.228], and grading instruct outputs −0.168 on [−0.228, −0.113]. Both
97 exclude zero by a wide margin and both point the same way.

98 The effect is an artifact of the marginals. Kappa depends on the marginal distributions as well as
99 on agreement, and the arms have very different ground truth base rates: the instruct generator is
100 correct on 77 percent of seen rows against 53 percent of unseen. An arm sitting near one half has far
101 more chance corrected headroom than a skewed arm, so the skew alone produces a gap. Feinstein
102 and Cicchetti [Feinstein and Cicchetti, 1990] describe this as the first of two kappa paradoxes, in
103 which high raw agreement is driven down by marginal imbalance. Figure 1 puts both statistics on
104 the same verdicts. Kappa separates the arms by 0.292; balanced accuracy separates them by 0.054
105 in the opposite direction.

106 Reporting this is not incidental. Chance corrected agreement is a standard choice for judge reliability,
107 and on a design where the arms differ in difficulty it will manufacture a membership effect from
108 nothing.

109 3.2 The dissociation

110 Table 1 and Figure 2 give the membership gap in balanced accuracy under both conditions. Supply-
111 ing the reference answer produces the behaviour a control should show: as judges improve the gap
112 contracts toward zero, from −0.054 to −0.036 to −0.023, with the strongest judge spanning zero.
113 On candidate solutions from the stage two generator the convergence is sharper, running −0.122,
114 then −0.002, then +0.008.

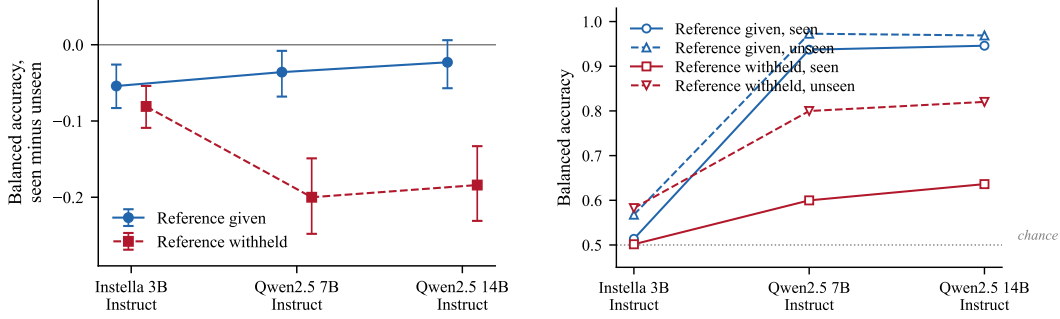


Figure 2: Left: membership gap in balanced accuracy across the judge ladder under both conditions, grading instruct outputs. Right: absolute balanced accuracy for all four cells of the design. The control condition confirms the two larger judges are competent instruments, reaching 0.94 to 0.97, so the reference condition free result is not read off a judge that cannot grade.

Table 2: Absolute balanced accuracy per cell, grading instruct outputs. Chance is 0.500.

Judge	Reference given		Reference withheld	
	Seen	Unseen	Seen	Unseen
Instella 3B Instruct	0.514	0.568	0.502	0.583
Qwen2.5 7B Instruct	0.937	0.973	0.600	0.800
Qwen2.5 14B Instruct	0.946	0.969	0.636	0.820

115 Withholding the reference removes the convergence. The gap grows once a judge is strong enough
116 to show it and then holds, at -0.081 , -0.200 and -0.184 . A defect of small models would shrink
117 as capability rises. This does the opposite between the first and second judge and then flattens.

118 3.3 Capability is demonstrated rather than assumed

119 A null or an effect from an incompetent judge says nothing, so absolute performance is reported
120 alongside the gaps. Table 2 gives every cell. Given the reference answer, the 7 and 14 billion
121 parameter judges reach 0.937 to 0.973, which is the regime in which their reference free numbers
122 become interpretable. The 3 billion parameter judge sits within 0.04 of chance on the seen arm in
123 every configuration, including when handed the answer, so its rows describe a noise floor and the
124 substantive claims rest on the larger two.

125 Three readings follow from the table taken together with Table 1. Capability is real, so conclusions
126 about the instrument come from instruments that work. The control behaves as a control, since
127 supplying the correct answer for the specific variant removes the gap, which is what a recall account
128 predicts and a pure difficulty account does not. And the reference free gap neither converges nor
129 responds to scale.

130 3.4 Abstention and what would falsify the account

131 A judge that declines to answer more often on one arm could produce the whole pattern with no
132 change in grading quality at all, since abstentions removed from the denominator reshape both sen-
133 sitivity and specificity. Abstention stays between 0.0 and 2.6 percent across every judge, condition
134 and arm, which is too small and too evenly spread to carry a gap of 0.200.

135 Three outcomes would have contradicted the account, and none occurred. Had the reference free
136 gap shrunk with capability in the way the control gap does, the effect would be a defect of weak
137 judges; instead it grows from the first judge to the second and then holds. Had the control gap failed
138 to converge, the arms would differ in some way that persists even when the answer is supplied,
139 pointing at difficulty rather than at recall; instead the strongest judge’s control interval spans zero.
140 And had the two larger judges failed to clear chance in the control condition, no reading of their
141 reference free numbers would be available at all; instead they reach 0.937 and 0.946 on the seen
142 arm.

143 4 Implications for evaluator construct validity

144 An evaluator is valid for a construct when its verdicts track that construct and nothing else. The
145 results here describe a specific and narrow failure of that condition. Under reference free grading
146 the measured quantity is not solely the quality of the reasoning presented, because it also varies with
147 whether the graded problem is one the evaluator is likely to have read. The variation is not small:
148 0.200 in balanced accuracy on the 7 billion parameter judge, against a control gap of 0.036 on the
149 identical items and the identical candidate solutions.

150 Two practical consequences follow. Reference free grading is the configuration used exactly where
151 a gold answer does not exist, which includes open ended generation, proof sketches, and any setting
152 where the task is to assess a chain of reasoning on its own terms, so the failure falls on the applica-
153 tions that most need a trustworthy judge. And a benchmark assembled from public sources is the
154 case where the affected items are most concentrated, since public availability is what puts an item
155 in a training corpus in the first place. Supplying a reference answer, where one exists, removes the
156 effect in these data, which makes it a mitigation rather than merely a diagnosis.

157 5 Mechanism

158 The pattern fits substitution of recall for verification. Asked to grade a problem present in its training
159 data without being told the answer, a judge appears to check the candidate solution against the
160 answer it remembers instead of against the reasoning in front of it. Two thirds of graded rows are
161 variants, and the answer changing variants have different correct answers from their originals, so a
162 remembered answer marks correct solutions wrong. Supplying the reference for the variant at hand
163 removes any need to recall, and the gap goes with it.

164 The account is inferential. Establishing it directly would require attributing a verdict to a specific
165 memorized document, which the present data cannot do.

166 6 Limitations

167 **Membership is verified for one judge, and it is not the judge carrying the result.** Containment
168 is measured against the Instella corpus. The Instella judge therefore has verified membership but
169 sits at chance. The Qwen judges show the effect, and their corpora are unknown, so for them the
170 partition is GSM8K train against test rather than verified seen against unseen. This asymmetry
171 is real and is the main reason the primary claim is stated as a within judge contrast between two
172 conditions over identical items and identical candidate solutions, which holds whatever the arm
173 label means for a given judge. What the Qwen results establish is that a competent judge grades this
174 partition differently depending on whether a reference is supplied. Closing the gap between that and
175 a membership claim requires a second open corpus model of comparable capability.

176 **One family supplies the candidate solutions.** Every graded output comes from an Instella check-
177 point. Whether the dissociation reproduces on solutions from other generators is untested.

178 **One checkpoint is unexplained.** The effect concentrates on stage two and instruct outputs. Stage
179 one outputs are wrong nearly everywhere and are trivially gradeable in either condition, which
180 accounts for that null. The supervised fine tuned checkpoint shows no gap in either condition despite
181 output quality comparable to instruct, and the present data do not explain why.

182 **Mechanism is inferred, and one prompt template is tested.** No attribution evidence links a
183 verdict to a training document, and robustness of the dissociation to prompt phrasing is untested.

184 7 Conclusion

185 Measured against membership fixed by containment rather than estimated, an LLM judge shows no
186 loss of reliability on problems from its training corpus when the reference answer is supplied, and a
187 substantial one when it is withheld. The gap under reference free grading runs to 0.200 in balanced
188 accuracy and does not close with scale, while the same judges in the control condition converge

189 toward zero as they improve. Chance corrected agreement inverts this picture on identical verdicts
 190 and should not be used where the arms differ in base rate. The operational reading is specific:
 191 reference free grading is the configuration deployed whenever no gold answer exists, and it is the
 192 configuration in which the items most likely to be in a judge’s training data are graded least reliably.

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228 **A Related work in full**

229 Zheng et al. [Zheng et al., 2023] introduce the LLM as judge paradigm at scale and document
230 position bias, verbosity bias and self preference, reporting agreement with human preference above
231 80 percent for strong judges. Their audits characterize the judge and the prompt; membership of the
232 graded item in the judge’s corpus is outside reach for the closed models they study, which is the gap
233 the present design addresses for one open family.

234 On the detection side, containment based overlap rules originate with Brown et al. [Brown et al.,
235 2020], and membership inference from model confidence is developed by Carlini et al. [Carlini
236 et al., 2023] and Shi et al. [Shi et al., 2024]. Yang et al. [Yang et al., 2023] show that rephrasing
237 defeats overlap based detection outright, which is why a study that needs a trustworthy treatment
238 label cannot rely on an estimated one.

239 Two lines of work establish the value of released intermediate checkpoints. Biderman et al. [Bider-
240 man et al., 2023b] release the Pythia suite specifically to make training dynamics analyzable, and
241 Biderman et al. [Biderman et al., 2023a] use such checkpoints to study when memorization emerges
242 during training. The present work applies the same logic to an evaluator rather than to a generator.

243 **B Containment method and calibration**

244 Item text is normalized to lowercase alphanumeric tokens and decomposed into every window of 13
245 consecutive tokens. Containment against a corpus is the fraction of an item’s windows found in it,
246 computed in one streaming pass with an inverted index from window to source item, so cost is linear
247 in corpus size and independent of how many item sets are indexed together. Because the corpus is
248 generated only from the GSM8K training split, every test item is a verified negative: the any match
249 rule flags 3 of 1000, a rate of 0.3 percent, and the adopted threshold flags none.

250 Identifier namespacing is load bearing. GSM8K train and test share an index scheme, so loading
251 both without namespacing collides items and produces containment values above unity, which is
252 impossible for a fraction. Anyone reproducing this design will meet the same trap.

253 **C Reproducibility controls**

254 Several controls are load bearing rather than incidental. The transformers version is pinned exactly,
255 since a version range places different checkpoints on different minor versions and that difference
256 enters the estimate as though it were a model difference. Batch size is fixed across all generation,
257 because greedy decoding in reduced precision is sensitive to padding and a batch size change alters
258 tie breaking. Variants are seeded per pair of item and perturbation type, so variant text is stable
259 regardless of call order and generations stay valid when the item set changes. Every generation
260 batch is flushed to disk and resume is keyed on item identifier, so an interrupted run continues
261 instead of restarting.

262 All generation records, containment values, judge verdicts and analysis code are released for audit at
263 a public repository and dataset mirror, both anonymized for review. Random seeds are fixed across
264 arm selection, variant construction and every bootstrap. An assistant helped implement generation,
265 containment and analysis code under the direction of the authors, who verified every reported num-
266 ber against the underlying verdict files; the assistant did not design the conditions or choose which
267 results to report.